LUCIDMSK CME PROGRAM

MODULE 12 — POST-TEST

Spine Trauma CT & MRI: Classification Systems & Instability

Spine · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 AO Spine thoracolumbar type B2 injury:

- **A** Compression fracture; posterior ligaments intact
- **B** Posterior tension band disruption — osseous (Chance fracture)
- **C** Posterior tension band disruption — ligamentous (posterior ligament complex intact on CT but disrupted on MRI)
- **D** Translation/rotation injury; complete instability

2 Denis three-column model: which column must be disrupted for a burst fracture to be unstable?

- **A** Anterior column alone
- **B** Anterior + middle columns
- **C** Middle + posterior columns
- **D** All three columns

3 Posterior Ligamentous Complex (PLC) is BEST assessed on which MRI sequence?

- **A** Sagittal T1 (non-fat-suppressed)
- **B** Sagittal STIR or T2 fat-saturation
- **C** Axial T2 gradient echo
- **D** Post-contrast T1 fat-sat

4 Thoracolumbar Injury Classification and Severity (TLICS) score ≥5 typically indicates:

- **A** Conservative management
- **B** Ambiguous — clinical decision required
- **C** Surgical intervention recommended
- **D** CT surveillance only

5 **Jefferson fracture (C1 atlas burst fracture) stability criterion on CT:**

A	Number of fracture lines
B	Total lateral mass overhang >6.9 mm (Spence rule) = transverse ligament rupture
C	C1-C2 interval measurement only
D	Number of fragments

6 **Hangman fracture (traumatic spondylolisthesis of C2) — Levine-Edwards Type II. Features:**

A	Bilateral C2 pedicle fractures; minimal displacement; stable
B	Bilateral C2 pedicle fractures; >3 mm displacement; disrupted C2-C3 disc; more unstable
C	Bilateral pedicle fractures + severe angulation + C2-C3 disc rupture; most unstable
D	Unilateral pedicle fracture only

7 **On MRI, anterior cord syndrome presents with:**

A	Posterior cord T2 signal change; position sense lost
B	Anterior cord T2 signal change; motor and pain/temperature lost; position sense PRESERVED
C	Complete motor and sensory loss below injury
D	Central cord T2 change; hands weakest; bladder dysfunction

8 **SLIC (Subaxial Cervical Injury Classification) scoring: which component carries the highest risk modifier for surgical decision?**

A	Injury morphology (compression = 1)
B	Discoligamentous complex (DLC) disruption (disrupted = 2)
C	Neurological status (intact = 0)
D	Distraction injury morphology (= 3)

9 **A thoracolumbar fracture shows anterior column compression + retropulsed middle column fragment in the canal at >50% stenosis. TLICS morphology score:**

- **A** 1 (compression)
- **B** 2 (burst)
- **C** 3 (distraction)
- **D** 3 (translation/rotation)

10 **Chance fracture MRI hallmark:**

- **A** Anterior column compression; posterior normal
- **B** Horizontal bone and/or soft tissue injury through ALL three columns in a flexion-distraction mechanism; STIR signal through ALL posterior elements
- **C** Posterior listhesis >3 mm without fracture
- **D** Central cord signal change; extremity weakness

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	C	B	C	B	B	B	B	B	B

Q1 **AO Spine thoracolumbar type B2 injury:** **Answer: B**

A Compression fracture; posterior ligaments intact

✓ **Posterior tension band disruption — osseous (Chance fracture)**

C Posterior tension band disruption — ligamentous (posterior ligament complex intact on CT but disrupted on MRI)

D Translation/rotation injury; complete instability

EXPLANATION AO Spine Type B2 = posterior tension band disruption through BONE (Chance fracture — transosseous). B1 = posterior ligamentous disruption (soft tissue, often missed on CT). B3 = anterior disruption in extension. C = translation/rotation.

Q2 **Denis three-column model: which column must be disrupted for a burst fracture to be unstable?** **Answer: C**

A Anterior column alone

B Anterior + middle columns

✓ **Middle + posterior columns**

D All three columns

EXPLANATION Denis burst fracture instability requires disruption of the MIDDLE column (posterior vertebral body/posterior longitudinal ligament) AND the POSTERIOR column. Middle alone = potentially unstable. Anterior alone = stable compression fracture.

Q3 **Posterior Ligamentous Complex (PLC) is BEST assessed on which MRI sequence?** **Answer: B**

A Sagittal T1 (non-fat-suppressed)

✓ **Sagittal STIR or T2 fat-saturation**

C Axial T2 gradient echo

D Post-contrast T1 fat-sat

EXPLANATION Sagittal STIR or T2 fat-sat is the most sensitive sequence for PLC disruption — showing hyperintense signal through the interspinous ligament, supraspinous ligament, ligamentum flavum, and facet joint capsule.

Q4 **Thoracolumbar Injury Classification and Severity (TLICS) score ≥5 typically indicates:** **Answer: C**

A Conservative management

B Ambiguous — clinical decision required

✓ **Surgical intervention recommended**

D CT surveillance only

EXPLANATION TLICS ≥5 = surgical recommendation. Score 4 = ambiguous/surgeon preference. ≤3 = conservative management. Components: injury morphology (1–3) + neurological status (0–3) + PLC integrity (0–3).

Q5 Jefferson fracture (C1 atlas burst fracture) stability criterion on CT: **Answer: B**

A Number of fracture lines

✓ **Total lateral mass overhang >6.9 mm (Spence rule) = transverse ligament rupture**

C C1-C2 interval measurement only

D Number of fragments

EXPLANATION Spence rule: total bilateral lateral mass overhang >6.9 mm on AP open-mouth view (or CT reconstruction) suggests transverse ligament rupture → unstable Jefferson fracture. Confirmed with dynamic X-rays or MRI of the transverse ligament.

Q6 Hangman fracture (traumatic spondylolisthesis of C2) — Levine-Edwards Type II. Features: **Answer: B**

A Bilateral C2 pedicle fractures; minimal displacement; stable

✓ **Bilateral C2 pedicle fractures; >3 mm displacement; disrupted C2-C3 disc; more unstable**

C Bilateral pedicle fractures + severe angulation + C2-C3 disc rupture; most unstable

D Unilateral pedicle fracture only

EXPLANATION Levine-Edwards Type II = bilateral C2 pedicle fractures with >3 mm displacement and disrupted C2-C3 disc, but without severe angulation. More unstable than Type I (<3 mm). Type IIA = minimal displacement but severe angulation. Type III = most severe (disc rupture + facet dislocation).

Q7 On MRI, anterior cord syndrome presents with: **Answer: B**

A Posterior cord T2 signal change; position sense lost

✓ **Anterior cord T2 signal change; motor and pain/temperature lost; position sense PRESERVED**

C Complete motor and sensory loss below injury

D Central cord T2 change; hands weakest; bladder dysfunction

EXPLANATION Anterior cord syndrome = anterior cord T2 signal change + loss of motor function AND pain/temperature sensation, with preserved posterior column function (proprioception/vibration/light touch). MOST severe incomplete cord injury.

Q8 SLIC (Subaxial Cervical Injury Classification) scoring: which component carries the highest risk modifier for surgical decision? **Answer: B**

A Injury morphology (compression = 1)

✓ **Discoligamentous complex (DLC) disruption (disrupted = 2)**

C Neurological status (intact = 0)

D Distraction injury morphology (= 3)

EXPLANATION DLC (Discoligamentous Complex) disruption is the highest single-component risk modifier — intact = 0, indeterminate = 1, disrupted = 2. Combined with neurological status (root injury = 1, cord incomplete = 2, cord complete = 3), drives surgical decision.

Q9 **A thoracolumbar fracture shows anterior column compression + retropulsed middle column fragment in the canal at >50% stenosis. TLICS morphology score:** **Answer: B**

A 1 (compression)

✓ **2 (burst)**

C 3 (distraction)

D 3 (translation/rotation)

EXPLANATION Burst fracture (middle column retropulsion into the canal) = TLICS morphology score 2. Compression = 1. Distraction = 3. Translation/rotation = 3 (same as distraction). Burst fractures can range TLICS 2–8 depending on neurological and PLC status.

Q10 **Chance fracture MRI hallmark:** **Answer: B**

A Anterior column compression; posterior normal

✓ **Horizontal bone and/or soft tissue injury through ALL three columns in a flexion-distraction mechanism; STIR signal through ALL posterior elements**

C Posterior listhesis >3 mm without fracture

D Central cord signal change; extremity weakness

EXPLANATION Chance fracture = horizontal transosseous (B2) or transligamentous (B1) injury through all three columns in a flexion-distraction mechanism. STIR shows high signal through interspinous/supraspinous ligaments, facets, and anterior anulus. Seatbelt injury pattern.